2 This question is about artificial satellites that orbit the Earth. You may treat the Earth as a perfect sphere with the centre of gravity at its centre. All other celestial bodies and their effects need not be considered.

The radius of the Earth is 6370 km and the mass of the Earth is 5.97×10^{24} kg.

(a) (i) Determine the orbital period of a geostationary satellite in seconds.

period = ... s [1]

(ii) Determine the orbital radius of a geostationary satellite to the centre of the Earth.

radius = ... m [2]

(b) The Molniya orbit (Figure 2.1) is a highly elliptical orbital path designed to provide continuous communication coverage to polar regions by positioning its furthest point directly over the North Pole.

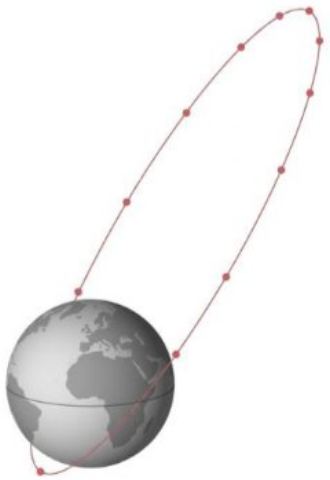


Figure 2.1

A satellite of mass 1500 kg is placed into this orbit. The closest distance the satellite approaches the Earth's centre is 6870 km while its furthest distance from the Earth's centre is 46200 km.

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(i) Calculate the change in gravitational potential experienced by the satellite as the satellite moves from the closest to the furthest points to Earth.

change in gravitational potential = ... J kg^{-1} [2]

(ii) Calculate the change in kinetic energy of the satellite as the satellite moves from the closest to the furthest points to Earth.

change in kinetic energy = ... J [2]

(iii) With reference to your answer in **(b)(ii)**, explain why designing the orbit to have its furthest point over the pole makes it highly effective for continuous communications to the polar region.

.....

.....

..... [1]

[Total: 8]